

Jacob Edwards, PhD

Data Scientist

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PhD scientist (Columbia, Auditory Neuroscience) building ML and analytics on TB-scale sensor data. A decade running controlled experiments, drawing inference from messy data, and turning quantitative results into clear visual stories. Created and taught Columbia's applied data science course with two student capstones published in peer-reviewed journals. Author of 6 data papers and NIH F31 awardee.

SKILLS

Languages & tools: Python (pandas, NumPy, scikit-learn, matplotlib), R (tidyverse, ggplot2), SQL, MATLAB, Git/GitHub, Jupyter

ML & statistics: Classification (SVM, Random Forest, ensemble methods), regression, time-series, NLP & sequence modeling, dimensionality reduction (PCA, wavelets), feature engineering, hyperparameter tuning, cross-validation

Experimentation: Experimental design, causal inference, hypothesis testing, A/B test methodology, bootstrapping, effect size estimation, simulation, metric design

Communication: Technical writing, data storytelling, stakeholder presentations, cross-functional collaboration, teaching & mentorship

EXPERIENCE

Research Scientist

Columbia University · New York, NY · Feb 2018 - Dec 2025

- Trained and deployed a 3-classifier ensemble (SVM, Random Forest, KDE) on **411M+ labeled observations across 155 GB** of time-series sensor data; eliminated hundreds of hours of manual quality review across 32 experiments and standardized labeling for 4 downstream research teams. Handled 93/7 class imbalance via balanced subsampling. ([write-up ↗](#))
- Built end-to-end ETL pipelines processing **3.2 TB of high-resolution sensor data** from 5,600+ channels across 20 subjects. Feature extraction, aggregation, validation, and visualization layered for downstream analyses.
- Shipped self-service interactive dashboards (3D, time-aligned audio + visual) so non-technical stakeholders could explore complex sensor data without engineering support; **adopted by 3 additional teams** as standard tooling.
- Designed and ran **controlled experiments with treatment/control manipulations** across 5 species to disentangle inherited vs. learned drivers of behavior. Results published in *Journal of Neuroscience* (2025) and *Scientific Reports* (in press, 2026).
- Built predictive sequence models (NLP-style; custom evaluation metric using fuzzy matching and bootstrapped confidence intervals) on **30k+ tokenized acoustic events from 180 subjects**; cluster-derived labels enabled supervised learning where ground truth was unavailable.
- Unified database and annotation infrastructure for 30k+ events / 180 subjects via custom GUI; **cut per-project labeling time ~80% across 3 teams**.

Statistics & Data Science Instructor

Columbia University · New York, NY · 2020 - 2024

Laboratory in Justice Data Science (2022, 2024)

Applied Data Science · [portfolio ↗](#)

- Designed and taught Columbia's original applied data science course covering R, data wrangling, visualization, ML, NLP, network analysis, and **algorithmic fairness** on real policy datasets (NYC OpenData, ProPublica COMPAS, NIJ recidivism). Wrote 20 hours of original lectures and 7 hands-on assignments.
- **Two student capstones published in peer-reviewed journals:** a geospatial analysis of police use-of-force and domestic violence in *Violence Against Women*, and a risk analysis of cybersecurity breaches and AI in the *Columbia Undergraduate Law Review*.
- Three students used the course as a launchpad to graduate programs at LSE, NYU (Urban Data Science), and Columbia's Center for Justice. Highly rated by students across both semesters.

Statistics for Behavioral Scientists (2020, 2022, 2024)

- Led lab sections teaching statistical reasoning and R to 40+ students per semester for 3 semesters.

Independent Data Science Work

Jan 2026 - Present

- Authored [Scientific Reports paper ↗](#) on a species-rules syntax model for sequential acoustic data; the approach generalizes to detecting and comparing sequential structure in complex audio signals, including speech.
- Built public [portfolio ↗](#) of ML, visualization, and instructional work with production write-ups and code.

SELECTED PROJECTS

SpikeJungle

MATLAB · ensemble methods · time-series

Production ML Classifier for Sensor Data · [portfolio ↗](#)

- 3-model ensemble (SVM, Random Forest, KDE) automating noise removal from voltage recordings; trained on 411M+ labeled observations across 155 GB. Validated on 32 unseen experiments with 74-92% inter-classifier agreement; preserved biological signal post-cleaning.

Species Syntax Model

R · MATLAB · NLP-style sequence modeling · bootstrapped evaluation

Sequence Modeling on Acoustic Data · [portfolio ↗](#)

- Predictive model that organizes acoustic units into sequences using species-level “grammar rules”, achieving sequence-prediction accuracy on par with tutor-based prediction. Published in *Scientific Reports* (2026); generalizes to any complex sequential audio signal.

SyllLabeler

R · MATLAB · GUI · clustering · dimensionality reduction

Annotation & Database Tool · [portfolio ↗](#)

- Unified interface for segmenting, labeling, and managing 30k+ acoustic events across 180 subjects; cut per-project labeling time ~80% across 3 teams.

EDUCATION

PhD, Psychology (Auditory Neuroscience)

Columbia University · 2025

MSc, Biology

Colorado State University · 2018

BSc, Biology

University of Tennessee, Knoxville · 2013

SELECTED PUBLICATIONS & RECOGNITION

- 6 peer-reviewed publications including first-author data papers in the *Journal of Neuroscience*, *Scientific Reports*, and *Journal of Comparative Neurology* · [Google Scholar ↗](#)
- \$150k+ in competitive research funding secured · NIH predoctoral grant (F31 [↗](#))
- Profiled by Columbia’s Zuckerman Institute [Science Life feature ↗](#)
- *Selected presentations*: Society for Neuroscience (2019, 2022), Taste of Science NYC (2021), Frontiers of Neuroscience, Barnard College (2021)